TUTOR SESSION ACTIVITIES:

OBJECT-ORIENTED PROGRAMMING

Activity 1 – Student Grader

A university lecturer wants to automate the process of calculating students' final marks for a Programming module. Currently, marks are calculated manually, which is time-consuming and prone to errors. To improve efficiency and accuracy, you have been tasked to develop a Java application using object-oriented principles.

Create a `Student` class that stores student details, `StudentID`, `StudentName`, and calculates the final mark based on the 3 assessments:

- Part 1 – contributes 25%
- Part 2 – contributes 30%
- Final POE (Portfolio of Evidence) – contributes 35%

The application then calculates the final mark using this formula:

Final Mark = (Part1 * 0.25) + (Part2 * 0.30) + (Part3 * 0.35)

The main program should:

- Allow the lecturer to enter a student's details and their 3 assessment scores
- Instantiate a `Student` object for the student. Use a parameterized constructor.
- Call a method inside the `Student` class that calculates the final mark using the provided formula
- Display:
    - the student details
    - individual assessment scores
    - the final mark

Activity 2 – Library Book Management System

You have been tasked to develop a Java application to manage books at the local library using a `Book` class.

When a librarian receives a new book donation, the system does not yet have its details stored.

The main program should:

- Instantiate a `Book` object using a default constructor.
- Prompts the librarian to enter:
  - Book title
  - Author name
  - ISBN number
- The entered values are assigned to the object using setter methods.
- Create four more `Book` objects using a parameterized constructor, passing all details directly during object creation.
- Display all five book records.